

Pokhara University
Faculty of Science and Technology

Course Code.: CMP 553 (3 Credits)
Course Title: Database Engineering (3-0-0)
Nature of the course: Theory & Research
Level: Master

Full marks: 100
Pass marks: 60
Total Lectures: 45 hrs
Program: MCE/MSc CS

1. Course Description

This course explores basic to advanced database engineering concepts beyond traditional relational models, covering modern architectures such as NoSQL, NewSQL, distributed databases, and big data technologies. It covers advanced indexing, query optimization, transaction management, and scalability techniques. This course expects to design and implement efficient real scenario database by providing essential knowledge and skills related to modern database and its application through case studies, group works, readings, and practical knowledge.

2. General Objectives

This course is designed to equip students with an overview of database, database management system, distributed database and its applications in various domains. It provides sufficient knowledge of analyzing advanced database architectures, including distributed and cloud-based systems, apply indexing and query optimization techniques for high-performance databases, Design and implement NoSQL databases (document, key-value, column-family, graph), Evaluate trade-offs in distributed databases (CAP theorem, consistency models), Utilize big data processing frameworks (Hadoop, Spark) in database applications, Develop scalable, fault-tolerant database solutions for real-world scenarios.

3. Contents in Detail

Specific objectives	Contents
Explain the basic database and database management system and data models	Unit I: Introduction (4 hrs) 1.1 Introduction to Database Systems 1.2 Database Management System 1.3 Data models 1.4 Relational Model and Entity-Relationship Diagrams
Describe the basic data modeling and fundamentals of SQL and queries.	Unit II: Data Modeling and SQL (4 hrs) 1.1 Concept of Relations 1.2 Referential integrity and foreign key 1.3 Relational algebra 1.4 Various types of joins 1.5 SQL Fundamentals: DDL, DML 1.6 Advanced SQL (Joins, Subqueries, Aggregation)
Implement query processing and its execution, compare different indexing strategies, analyze transaction management in stored procedures.	Unit III: Advanced Relational Databases (5 hrs) 3.1 Query Processing and Optimization 3.2 Advanced Indexing (B trees, Hash Indexing, Bitmap Indexing) 3.3 Materialized Views and Caching Strategies 3.4 Stored Procedures, Triggers, and UDFs

• Apply fragmentation and replication strategies effectively. • Analyze CAP/PACELC trade-offs in system design. • Choose appropriate consistency models based on application needs. • Implement distributed transactions using 2PC, 3PC, Paxos, or Raft.	Unit IV: Distributed Database Systems (8 hrs) 4.1 Architectural models for distributed DBMSs - Client/Server systems - Peer-to-Peer systems - Multidatabase systems 4.2 Distributed Database Architectures (Fragmentation, Replication) 4.3 CAP Theorem (Consistency, Availability and Partition Tolerance) and PACELC Trade-offs 4.4 Consistency Models (Strong, Eventual, Causal) 4.5 Distributed Transactions (2PC, 3PC, Paxos, Raft) 4.6 Security in Distributed Database (Authentication, Authorization, Encryption, Integrity)
• Select appropriate NoSQL databases for different application needs • Design efficient NoSQL data models, Perform CRUD operations in MongoDB with aggregation and sharding • Configure Cassandra clusters with tunable consistency and Query and analyze graph data using Neo4j and Cypher	Unit V: NoSQL Databases (7 hrs) 5.1 Types of NoSQL Databases (Document, Key-Value, Column-family, Graph) 5.2 NoSQL data Models 5.3 CRUD, Aggregation, Sharding operation and its implementation using NoSQL database tools (MongoDB, Cassandra etc) 5.4 Graph Traversal and Graph databases and its implementation
• Process petabyte-scale data using Hadoop/Spark • Choose appropriate time-series databases for IoT/telemetry • Implement blockchain-based data storage solutions • Compare modern data architectures for different use cases	Unit VI: Big Data & Modern Database Technologies (7 hrs) 6.1 Hadoop Ecosystem (HDFS, MapReduce) 6.2 Apache Spark (RDDs, DataFrames, Spark SQL) 6.3 Time-Series Databases and its implementation 6.4 Blockchain Databases (Decentralized Storage)
• Optimize database performance through partitioning and indexing • Reduce latency with intelligent caching strategies • Design highly available database architectures • Leverage cloud-managed databases for scalability and reliability	Unit VII: Database Scalability and Optimization (6 hrs) 7.1 Database Partitioning (Horizontal vs. Vertical) 7.2 Caching Strategies (Redis, Memcached) 7.3 Load Balancing and High Availability 7.4 Indexing Strategies (Local, Global, Secondary) 7.5 Monitoring and Performance Testing

- Describe emerging and future trends in database, their issues, implementation and applications

Unit VIII: Emerging Trends

- 8.1 Cloud Database
- 8.2 Serverless Databases
- 8.3 Vector Databases for AI/ML
- 8.4 Edge Databases (IoT Applications)
- 8.5 Ethical Considerations in Database Engineering

5. Case Studies:

Case Studies shall cover the following works/studies with suitable tools like MySQL, PostgreSQL, PostGIS, MongoDB, Python, Grafana, AWS RDS, Azure Cosmos DB etc: SQL & Advanced SQL operations, Query Optimization and Execution Plans, CRUD Operations & Indexing, Distributed Transactions (2PC Simulation), Time Series Data, Spatial Data, Security issues in distributed databases.

6. Evaluation system and Students' Responsibilities

Evaluation System

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, case studies, research paper review, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

External Evaluation	Marks	Internal Evaluation	Marks
Semester-End Examination	40	Class attendance & participation	5
		assignments & class test	10
		Case Studies/ Research Paper Review	15
		Internal Term Exam	30
Total External	40	Total Internal	60
Full Marks 40+6 100			

Students' Responsibilities:

Each student must secure at least 60% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such a score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there will not be any provision for a re-exam.

7 Prescribed Books and References

Textbooks

- Database System Concepts, Seventh Edition, Avi Silberschatz, Henry F. Korth, S. Sudarshan, 2011, McGraw-Hill, ISBN 978-0-07-352332-3

2. Designing Data-Intensive Applications, Martin Kleppmann, O'Reilly Media, Inc., 2017, ISBN: 9781491903100.
3. "NoSQL Distilled" – Pramod J. Sadalage, Martin Fowler

Reference Books

1. Principles of Distributed Database Systems, M. Tamer Özsu, University of Waterloo and Patrick Valduriez University of Montpellier, 4th Edition, Springer Nature Switzerland AG 2020, Hardcover ISBN 978-3-030-26252-5.
2. PostgreSQL Internals, Egor Rogov, Postgres Professional, 2022–2023, ISBN 978-5-6045970-4-0.
3. Database Internals: A Deep Dive into How Distributed Data Systems Work, Alex Petrov, First Edition, O'Reilly, ISBN-13 978-1492040347, 2019